LAM BAO TRAN

lambaotran309@gmail.com | github.com/Baaotraan0407 | Baaotraan0407.github.io

PROFILE

Informatics graduate (GPA 3.81/4.00) with a prior degree in Finance and Banking, applying to graduate study in Industrial Engineering and Management. Interested in applying data analytics, optimisation and computer vision to sustainable operations, process improvement and evidence-based operational decisions.

EDUCATION

B.S. in Informatics, International Bachelor Program in Informatics (IBPI) 2022 – 2026
Yuan Ze University, Taiwan · GPA 3.81/4.00

B.A. in Finance and Banking, Corporate Finance 2016 – 2020
University of Economics and Finance, Ho Chi Minh City, Vietnam · GPA 3.34/4.00

SELECTED COURSEWORK — YUAN ZE UNIVERSITY

Mathematics: Calculus II (100), Discrete Mathematics (100), Calculus I (97), Linear Algebra (89)
Data and manufacturing: Databases (95), Big Data (93), Image Processing (93), Data Mining (91), Smart Manufacturing (86)

RESEARCH

Do Hourly Carbon Signals Overstate Scope 2 Reductions? Average vs. Marginal Emission Factors in Carbon-Aware Production Scheduling with Taiwan Grid Data 2026
Under review, 2026 International Conference on New Management Challenges in the ESG Era
First author, with Nguyen Ha Phuong Du and Dam My Tran

- Estimated hourly average and marginal emission factors for Taiwan's grid from public data.
- Scheduled 13 flow shops with constraint programming (OR-Tools CP-SAT) against a time-of-use cost baseline.
- Found average factors report 1.8–2.8× the marginal CO_2 cut (coal-heavy grid); most savings came from idle energy.

ACADEMIC PROJECTS

Real-Time Driver Drowsiness Detection — Team project, extended individually
Code: github.com/Baaotraan0407/drowsiness-detection | Demo: youtube.com/watch?v=mZabZnndXGw

- With a teammate: built a YOLOv8 + Haar + CNN pipeline with PERCLOS-based alerts.
- Individually: fixed the evaluation (subject-level split); eye CNN 95.8% closed-eye recall on unseen people.
- Error analysis: unlike eye closure, yawning differs from talking only over time, and the Haar mouth ratio is constant by design; added temporal smoothing that made live yawn detection work.
- Real-time on a laptop CPU (20 FPS; 26 with GPU) and an eye-closure alert at 1.5 s instead of about 3 s.

TrueCareer — Career Guidance Platform — Group project, Business Analyst
Code: github.com/Baaotraan0407/truecareer-career-guidance-system

- Wrote user stories (business rules, acceptance criteria) and built a high-fidelity UI prototype in Axure.
- Built an exploratory K-Means prototype on 800 synthetic profiles to test recommendation logic.
- Tested clusters (silhouette ≈ 0.10); showed the high ARI (0.85) only reflects how the synthetic data was made.

AWARD

Honorable Mention (team), IBPI Special Project Exhibition (drowsiness detection) Dec 2025

EXPERIENCE

Business Operations Assistant 2021 – 2022
Dien Thang Manufacturing & Trading One Member Co., Ltd., Ho Chi Minh City

- Coordinated goods receiving and dispatch for copper wires and electrical cables.
- Supported quotation preparation and customer pricing communication.

SME Credit Intern 2020 (3 months)
Vietnam Prosperity Joint Stock Commercial Bank (VPBank), SME Center, Gia Dinh Branch, Ho Chi Minh City

- Supported credit staff with SME loan documentation and lending procedures.
- Reviewed business and financial information; wrote an internship report on SME lending workflow.

Part-time Assistant, *Korea Trade-Investment Promotion Agency (KOTRA)* Jun 2018

SKILLS & LANGUAGES

Programming and tools: Python, C++, HTML/CSS, JavaScript; Git, GitHub Actions, VS Code, Jira, Streamlit
AI, data and optimisation: TensorFlow, Keras, YOLOv8, OpenCV, scikit-learn; OR-Tools CP-SAT
Languages: Vietnamese (native) · English (IELTS 6.0) · Chinese (TOCFL B1)